

Supervisor brief — progress since 18 June

Date: 23 June 2026 · **Period covered:** 19–21 June 2026 · **Project:** *The Geometry of Machine Reasoning*
MSc Computational Finance · joint industry project with Holistic AI · supervised by Dr Paolo Barucca (UCL) & Zekun Wu (Holistic AI)

At the 18 June meeting the cleaned geometry was citable for the first time, but three things were still open: the annotator-robustness loop was *running, not closed*; Phase-7 steering was *built but unrun*; and we hadn't settled *which layer* to steer at. The last few days **closed the robustness loop (2-way), turned the layer question into a clean methodological result, and hardened the Phase-7 design against a critique** before spending GPU on it. One new strand opened: finance as a reasoning domain.

Headlines

What happened	Detail	Why it matters
1. Annotator-robustness loop closed (2-way)	The geometry replicates across annotators : intrinsic dimension <i>and</i> the curvature pattern reproduce on Sonnet-4.5 ↔ Qwen3-235B labels (corr-dim ~6–8 in both, same ordering, low-dim keystone passes in both, curvature-as-chain-artefact holds) — despite κ being only 0.44 .	The external-validity result the robustness argument was waiting on. The manifold is a property of the <i>model</i> , not of one labeller.
2. Layer selection: confounded → redesigned → de-confounded	The attribution-patching method (E3) ran and was confounded — every behaviour ramps monotonically to L26–27 (a read-out-proximity artefact), giving no per-behaviour layer. Rebuilt as a <i>forward-intervention</i> steering-effect sweep with a random-direction null, per-layer α -normalisation and a bootstrap shortlist. De-confounded test: backtracking's causal layer is L16 (mid-network), not L27 — matches Venhoff (~L15–18) . Full sweep now running on the cluster.	This was the hidden blocker under "run Phase-7": we now have a principled, literature-consistent way to choose the steering layer.
3. Phase-7 steering hardened after adversarial critique	A methodology-evaluation pass judged the original design " <i>not fundable as-is</i> ." Added: a k-sweep; a random-subspace control; an energy-matched random baseline (the old norm-matched one was ~19× too weak — a real flaw caught); an orthogonal-complement arm; and a proper stats module (paired BCa bootstrap + Holm).	The causal headline experiment is now de-risked and worth the GPU spend — ready to run rather than likely to give an un-defensible result.

Supporting & smaller items

- **No sub-types, confirmed at the new layers.** Re-ran clustering at the reconciled layers (16/16/16/12); silhouette stays **0.18–0.20** — each behaviour is one continuous region, robust to layer choice.
- **Pooling question settled.** Our mean-pooling matches Venhoff's published recipe (verified in their code) — one less methodological wobble.

- **Gate-0 results ledger created** (RESULTS_LEDGER.md) as the canonical "what we found + trust status" tracker; candidate steering layers reconciled to 16/16/16/12.
- **New strand — finance as a reasoning domain (UCL).** Built a concrete-metric quant-exam benchmark (200 items: mental arithmetic, expected value, risk/sizing, no-arbitrage pricing, estimation) with a scoring harness that **rewards calibrated abstention over wrong answers**. An E0 pilot validated the harness end-to-end with oracle solvers (perfect → **+1.0**, wrong → **-1.0**, abstain → **0.0**). Scaffolding only — no model runs yet.

Status of the decisions raised on 18 June

Decision	Status	Where it stands now
#2 — Run Phase-7 steering (the causal Movement-I headline)	DE-RISKED	Not just "ready" — hardened after the critique; still needs your go-ahead + cluster GPU. The layer-selection blocker is now resolved.
#1 — Scale correctness labels 183 → ~1000 (predictive geometry)	PENDING	Still your call (~5× judge spend). No change since 18 June.
Layer-selection method (implicit blocker under #2)	RESOLVED	Forward-intervention sweep; backtracking → L16, consistent with Venhoff.

Still open / owed

- **UNRUN** **Phase-7 steering** — the causal Movement-I result. Awaiting go-ahead + GPU.
- **PARTIAL** **R2.2 third arm (Nova-Pro)** and the variance-ratio specificity-null replication are still pending; the 2-way verdict stands now.
- **CAVEAT** The Qwen3 arm ran on lossy analysis-side dedup (dup 33–52%), not a clean re-extraction (Sonnet ~1%) — so the replication holds *in the pattern*, not in the exact absolute numbers. To state plainly.
- **BLOCKED** **Predictive-geometry label scaling** — gated on decision #1.

What I'd value from you

- **Go / no-go on Phase-7** now that it's hardened — this is the missing *causal* leg of Movement I.
- **The label-scaling spend** (#1) for predictive geometry — still open.
- **A steer on the finance strand:** how far to push it as a parallel reasoning domain vs. keeping it a contained UCL-facing chapter.

Source trackers: RESULTS_LEDGER.md (what/status), CONFOUNDINGS_AND_REMEDIATION.md (why-wrong), METHODOLOGY.md (how). Builds on the 18 June brief (results/supervisor_meeting/MEETING_2026-06-18.md). Numbers as of the clean Gate-0 re-run; cluster layer-sweep + R2.2 Nova-Pro arm still in flight.